PARTH BALAR

Site Reliability Engineer | Kubernetes • Terraform • AWS • Observability • SLO/SLI
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TECHNICAL SKILLS

Cloud & Containerization: AWS (EC2, EKS, RDS, VPC, IAM, CloudWatch, S3), GCP, Kubernetes, Docker, Microservices
Infrastructure as Code: Terraform (modules: EKS, RDS, VPC, IAM), Ansible, Reusable IaC Modules
Observability & Monitoring: Prometheus, Grafana, Splunk (Cloud & Enterprise), CloudWatch, ELK Stack, PagerDuty, Real-time Alerting, Dashboards
SRE Principles: SLA, SLO, SLI, Error Budgets, Postmortems, Reliability-focused System Design, Incident Response, On-call, Toil Reduction
CI/CD & DevOps Tools: GitHub Actions, Jenkins, GitLab CI/CD, Git, Blue-Green & Canary Deployments, Container Orchestration
Programming & Scripting: Python (primary), Bash, Shell Scripting, SQL, JavaScript, Java, C/C++
Networking & Linux: Linux/Unix administration, HTTP/HTTPS, SSL/TLS, DNS, TCP/IP, Load Balancing
Databases, AI Tooling & Methods: PostgreSQL, MySQL, MongoDB, SQLite · LangChain, ChromaDB, RAG · Agile, Cross-functional Collaboration, Mentoring

PROFESSIONAL EXPERIENCE

DevOps / SRE Engineer | Citta Solutions Jun 2024 – Present
Plano, TX

- Drive reliability and performance of production services on AWS EKS Kubernetes by defining SLOs/SLIs, improving observability through Prometheus, Grafana, and CloudWatch, and proactively identifying system bottlenecks across cloud environments.
- Automate infrastructure and operations using Terraform, Kubernetes, and CI/CD tools (GitHub Actions) to eliminate toil and enable scalable, fault-tolerant deployments via blue-green and canary strategies, cutting release cycle time by 35%.
- Build monitoring systems with Prometheus, Grafana, and Splunk for real-time alerting and root-cause analysis; developed an LLM-powered on-call assistant (LangChain + Claude API) that triages PagerDuty alerts and queries production context, cutting manual triage time by 50% and MTTR by 45%.
- Lead incident management through on-call rotations, conducting blameless postmortems, and implementing automated recovery runbooks via ChromaDB-backed RAG retrieval to minimize downtime.
- Collaborate cross-functionally with product, infrastructure, and DevOps teams to integrate ML-based anomaly detection into deployment gates, reducing incidents and ensuring architectural clarity.

Site Reliability Engineer (Contract) | Splunk Jun 2022 – Dec 2023
Ahmedabad, India

- Owned end-to-end SLA/SLO/error-budget program for Splunk Cloud across distributed Kubernetes services on AWS and GCP, maintaining 99.9% uptime through Prometheus/Grafana alerting and Python/Bash automated remediation.
- Built CI/CD pipelines with Jenkins, Git, and Terraform for zero-downtime blue-green releases of containerized services on Kubernetes, cutting deployment cycle time by 40%.
- Authored reusable Terraform IaC modules adopted as team standard for new service onboarding, cutting provisioning time by 60% and reducing toil across Linux-based environments fleet-wide.
- Re-architected indexing/ingestion pipelines with HTTP/HTTPS/TLS endpoints, reducing query latency by 30% and ingestion overhead by 20%; surfaced regressions via Splunk and Grafana dashboards before user impact.
- Led blameless postmortems and chaos-engineering reviews; mentored junior engineers on SRE best practices and converted findings into automated recovery runbooks, cutting MTTR by 40%.

Site Reliability Engineer Intern | Crest Data Dec 2021 – May 2022
Ahmedabad, India

- Containerized legacy services with Docker and deployed on Kubernetes (AWS); automated provisioning via Terraform and Jenkins, reducing deployment setup from hours to minutes.
- Implemented SLI/SLO observability using Grafana and Splunk and introduced postmortem culture from scratch, contributing to a 35% reduction in MTTR within the first quarter.
- Developed an internal Python (FastAPI) + React automation tool replacing manual data-processing workflows, delivering 60% reduction in manual effort and 40% fewer data errors.

PROJECTS

FRIDAY — Containerized AI Assistant Platform - *Python, FastAPI, LangGraph, Docker Compose, Nginx, ChromaDB*

- Designed a containerized full-stack platform (Python 3.13 + FastAPI + React 18) deployed via Docker Compose with Nginx reverse proxy and HTTPS/SSL-terminated routing, supporting 112+ tools across 36 modules.
- Built a LangGraph ReAct agent with a two-tier intent router (regex <1ms + LLM fallback) handling 55+ intents across email, file I/O, browser automation, and system control.
- Implemented semantic memory with ChromaDB and a Markov-chain predictive prefetch engine; built a 5-source parallel pipeline using ThreadPoolExecutor with SQLite tracking.

Graph-Enhanced RAG Retrieval System - *Python, sentence-transformers, ChromaDB, BFS, Cosine Similarity*

- Built a hybrid retrieval pipeline combining dense vector search (all-MiniLM-L6-v2 embeddings) with BFS knowledge-graph traversal to surface structurally related code chunks invisible to keyword search.
- Pre-computed L2-normalised entity embeddings enabling single matrix-multiply similarity scoring across 80+ entities; merged graph and vector results for up to 50 deduplicated hits per query.

EDUCATION

M.S. Information Technology Management | The University of Texas at Dallas Aug 2023 – May 2025
B.S. Information Technology | Dharmsinh Desai University, India Jun 2018 – May 2022